



nopCommerce

Observability in the Wild

Presented by:

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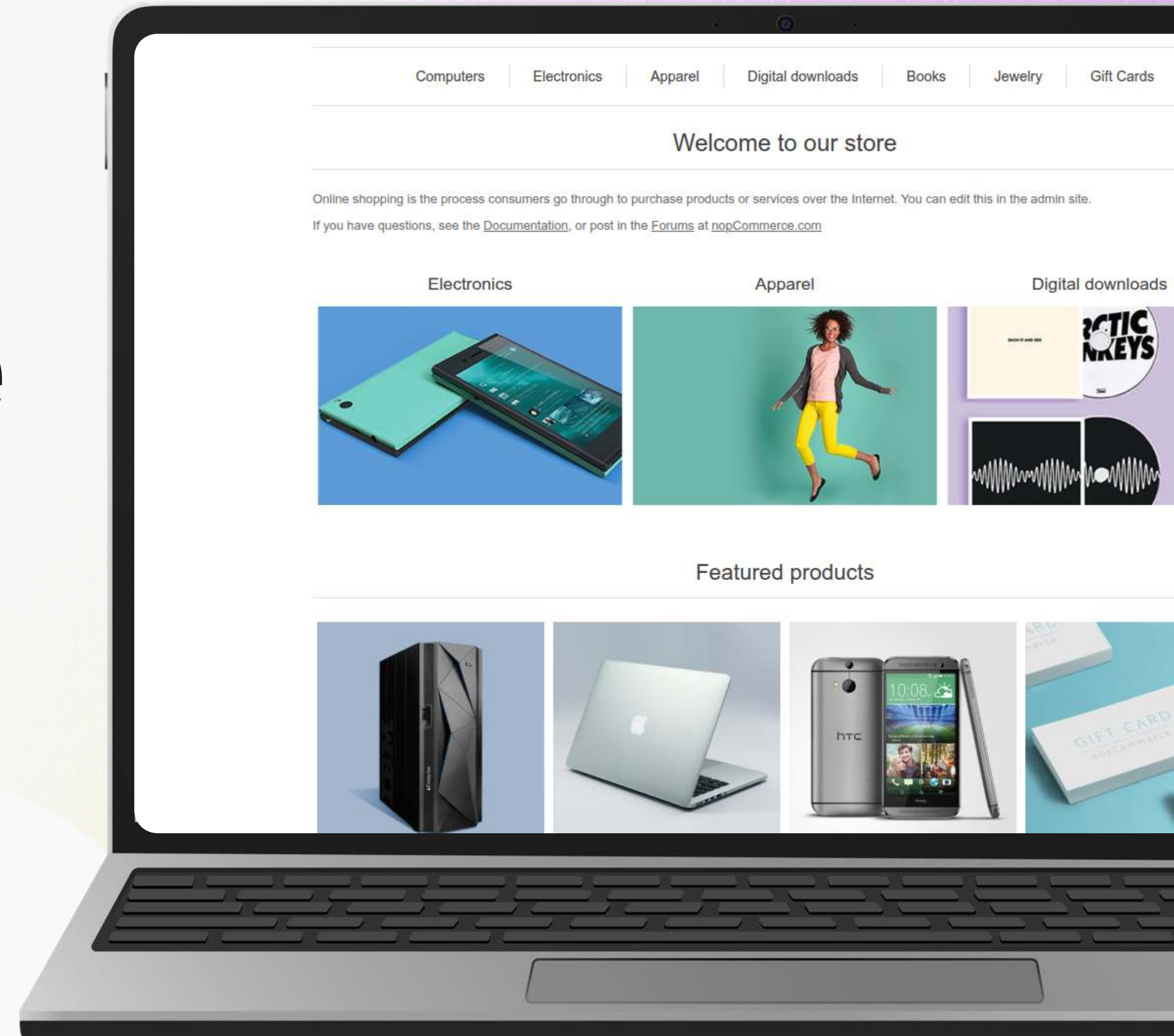
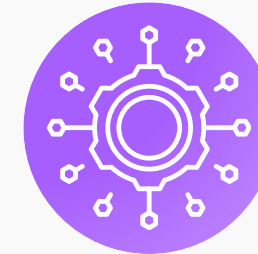


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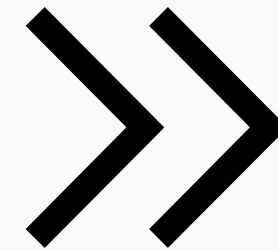
**CRITIQUE AND
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Problem & Goal

- nopCommerce has a **layered architecture** and a real checkout flow, but its operational **behaviour is hard to observe end-to-end**.
- Without focused instrumentation, failures in basket, inventory, payment or latency are difficult to localise under load.



Instrument the checkout path end-to-end with OpenTelemetry, producing actionable metrics and traces with a boundary-first, low-risk approach and no PII in telemetry.



USER FLOW SELECTED

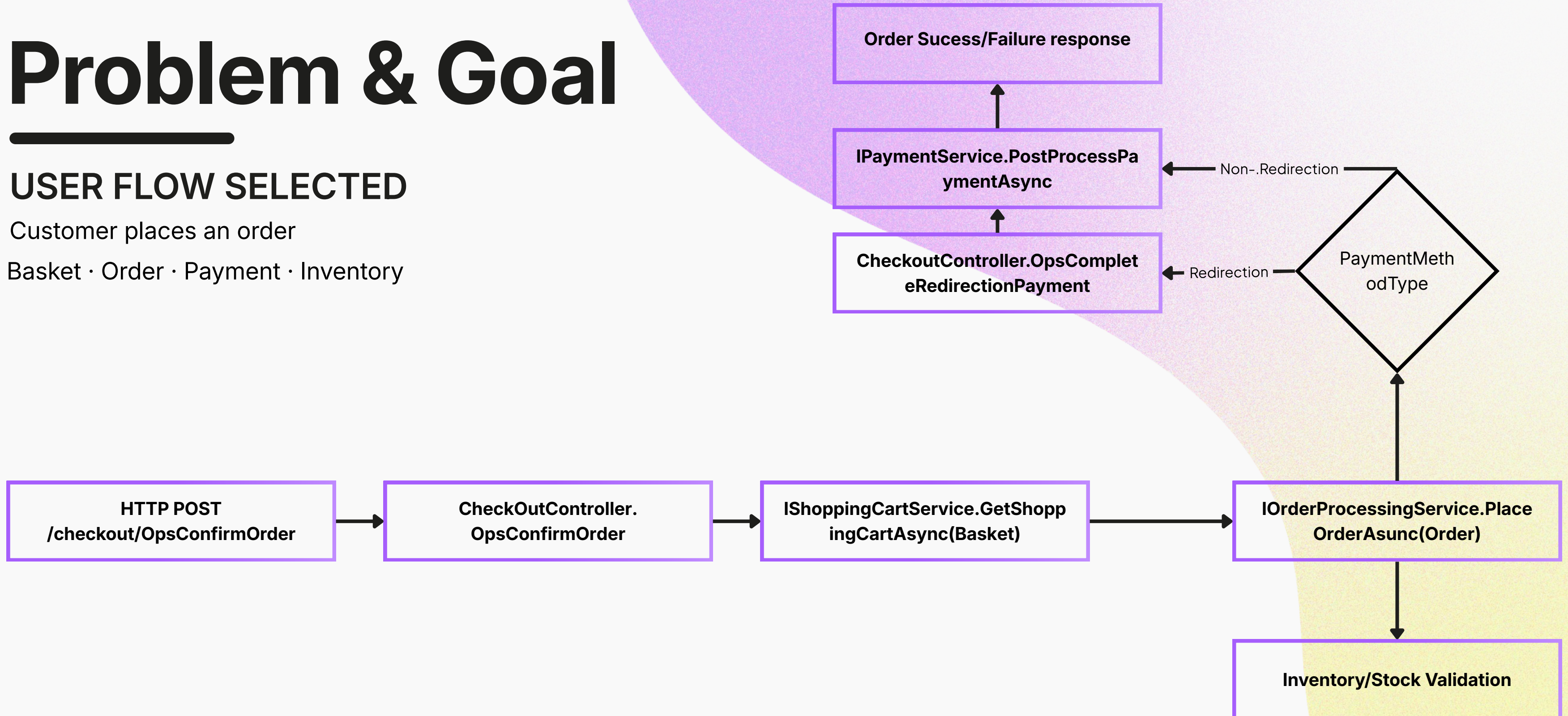
Customer places an order
Basket · Order · Payment · Inventory

Problem & Goal

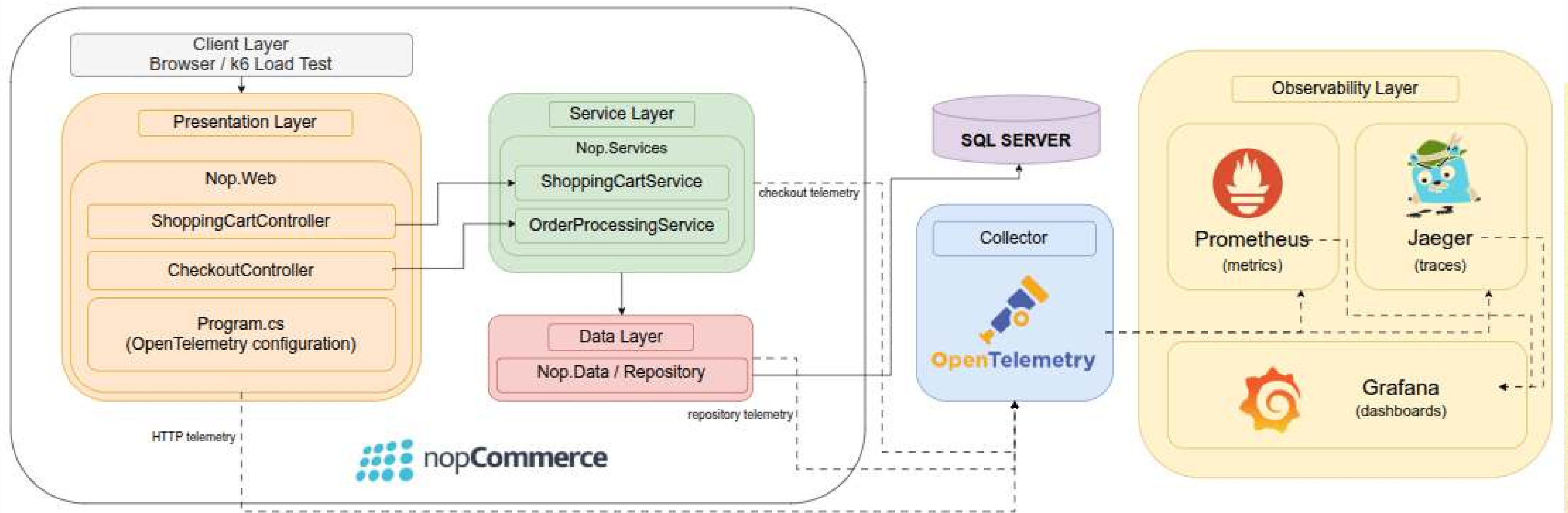
USER FLOW SELECTED

Customer places an order

Basket · Order · Payment · Inventory



Architecture



INSTRUMENTATION DECISIONS

Spans

HTTP POST /checkout/OpcConfirmOrder

Root span for the checkout request

checkout.place_order

Order placement orchestration

checkout.payment.post_process

Payment processing boundary

checkout.payment.redirection_complete

Redirect payment completion

event.publish

Internal event dispatch boundary

event.consume

Per-consumer event execution and fan-out cost

Key Metrics

checkout_attempts_total

Counter: checkout attempts split by success and failure

checkout_duration_seconds

Histogram: average and p95 checkout duration

checkout_step_dropoff_total

Counter: drop-off at critical checkout steps

basket_checkout_failures_total

Counter: basket failures by reason

inventory_checkout_failures_total

Counter: inventory failures by reason

DESIGN CHOICES AND PRIVACY TRADE-OFFS

Boundary-first approach

Telemetry was added at stable boundaries: CheckoutController, PlaceOrderAsync, payment processing, and IEventPublisher.

Alternative avoided

Broad tracing inside business logic was avoided to reduce coupling, code spread, and regression risk.

Privacy and cardinality

Sensitive attributes were removed before export, and only low-cardinality operational labels were used.

Architectural trade-off

The solution favors maintainability and actionable production signals over deep method-level tracing.

DEMO

- Grafana
Real-time dashboard for checkout health
- Prometheus
Metric queries and validation
- Jaeger
Trace-level inspection of checkout flows



CRITIQUE - CONCLUSION

What Helped

Clear layered architecture
Stable runtime entry points in controllers
Central OpenTelemetry setup in Program.cs
Existing service and repository boundaries



What Was Challenging

Multiple checkout paths
Partial-success and domain-level failures
Environment-dependent payment behaviour
Trace filtering still depends on runtime tagging quality

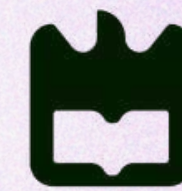


A boundary-first, low-risk instrumentation strategy was enough to turn a real checkout flow into an observable business path with actionable metrics and traces.

THANK YOU nopCommerce QUESTIONS?

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Online shopping is the process consumers go through to purchase products or services over the Internet. You can edit this in the admin site.

If you have questions, see the [Documentation](#), or post in the [Forums](#) at [nopCommerce.com](#)

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